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Setting GUIDs and MACs

To set GUIDs/MACs/UID for the given device, use the 'sg' (set guides) command with the -guid(s), -uid and/or -mac(s) flags.

4th Generation (Group I) Devices

On 4th generation/Group I devices, the "sg" command can operate on both the image file and the image on the flash. When running the "sg" command on an image on the flash, if the GUIDs/MACs/UIDs in the image are non-blank, the flint will re-burn the current image using the given GUIDs/MACs/UIDs.

1. Change the GUIDs/MACs on a device:

```
# flint -d /dev/mst/mt4099_pci_cr0 q -W- Running quick query - Skipping
full image integrity checks. Image type: FS2 FW Version: 2.42.5000 FW
Release Date: 4.5.2017 Device ID: 4099 Description: Node Port1 Port2
Sys image GUIDs: f45214030001b8a0 f45214030001b8a1
f45214030001b8a2 f45214030001b8a3 MACs: f4521401b8a1
f4521401b8a2 VSD: PSID: MT_1090120019 # flint -d
/dev/mst/mt4099_pci_cr0 -guid 0x452140300abadaba -mac
0x300abadaba sg -W- GUIDs are already set, re-burning image with the
new GUIDs ... You are about to change the Guids/Macs/Uids on the
device: New Values Current Values Node GUID: 452140300abadaba
f45214030001b8a0 Port1 GUID: 452140300abadabb
f45214030001b8a1 Port2 GUID: 452140300abadabc
f45214030001b8a2 Sys.Image GUID: 452140300abadabd
f45214030001b8a3 Port1 MAC: 00300abadaba f4521401b8a1 Port2
MAC: 00300abadabb f4521401b8a2 Do you want to continue ? (y/n) [n] :
y Burning FS2 FW image without signatures - OK Restoring signature -
OK # flint -d /dev/mst/mt4099_pci_cr0 q Image type: FS2 FW Version:
2.31.5050 FW Release Date: 4.5.2014 Device ID: 4099 Description: Node
Port1 Port2 Sys image GUIDs: 452140300abadaba 452140300abadabb
452140300abadabc 452140300abadabd MACs: 00300abadaba
00300abadabb VSD: PSID: MT_1090120019
```

2. Change the GUIDs/MACs on an image file:

```
# flint -i /tmp/image.bin q Image type: fs2 FW Version: 2.31.5050 FW
Release Date: 4.5.2014 Device ID: 4099 Description: Node Port1 Port2
Sys image GUIDs: f45214030001b8a0 f45214030001b8a1
f45214030001b8a2 f45214030001b8a3 MACs: 00300abadaba
00300abadabb VSD: PSID: MT_1090120019 # flint -i /tmp/image.bin -
guid 0002c9000abcdef0 -mac 02c90abcdef0 sg You are about to
change the Guids/Macs/Uids on the device: New Values Current Values
Node GUID: 0002c9000abcdef0 f45214030001b8a0 Port1 GUID:
0002c9000abcdef1 f45214030001b8a1 Port2 GUID: 0002c9000abcdef2
f45214030001b8a2 Sys.Image GUID: 0002c9000abcdef3
f45214030001b8a3 Port1 MAC: 02c90abcdef0 00300abadaba Port2
MAC: 02c90abcdef1 00300abadabb Do you want to continue ? (y/n) [n] :
y Restoring signature - OK# flint -i /tmp/image.bin q Image type: FS2 FW
Version: 2.31.5050 FW Release Date: 4.5.2014 Device ID: 4099
Description: Node Port1 Port2 Sys image GUIDs: 0002c9000abcdef0
0002c9000abcdef1 0002c9000abcdef2 0002c9000abcdef3 MACs:
02c90abcdef0 02c90abcdef1 VSD: PSID: MT_1090120019
```

5th Generation (Group II) Devices

On 5th Generation (Group II) devices, the “sg” command can operate on both the image file and the image on the flash. When running the “sg” command on an image on the flash, -uid flag must be specified. For ConnectX-4, -guid/-mac flags can be specified. By default, 8 GUIDs will be assigned for each port starting from base, base+1 up until base+7 for port 1 and base+8 up until base+15 for port 2.

To change the step size and the number of GUIDs per port, specify `guids_num=<num> step_size=<size>` to the sg command.

1. Change GUIDs for device:

```
# flint -d /dev/mst/mt4113_pciconf0 q Image type: FS3 FW Version:
10.10.3000 FW Release Date: 29.4.2014 Description: UID GuidsNumber
Step Base GUID1: 0002c903002ef500 8 1 Base GUID2:
0002c903002ef508 8 1 Base MAC1: 0002c92ef500 8 1 Base MAC2:
0002c92ef508 8 1 Image VSD: Device VSD: VSD PSID: MT_1240110019
# flint -d /dev/mst/mt4113_pciconf0 -uid 0002c123456abcd -ocr sg -W-
Firmware flash cache access is enabled. Running in this mode may cause
```

the firmware to hang. Updating GUID section - OK Updating ITOC section
- OK Restoring signature - OK # flint -d /dev/mst/mt4113_pciconf0 q
Image type: FS3 FW Version: 10.10.3000 FW Release Date: 29.4.2014
Description: UID GuidNumber Step Base GUID1: 00002c123456abcd 8 1
Orig Base GUID1: 0002c903002ef500 8 1 Base GUID2:
00002c123456abd5 8 1 Orig Base GUID2: 0002c903002ef508 8 1 Base
MAC1: 00002c56abcd 8 1 Orig Base MAC1: 0002c92ef500 8 1 Base
MAC2: 00002c56abd5 8 1 Orig Base MAC2: 0002c92ef508 8 1 Image
VSD: Device VSD: VSD PSID: MT_1240110019

Orig Base GUID/MAC refers to the GUIDs/MACs located in the
MFG(manufacture guids) section of the flash/image.

2. Change GUIDS for device (specifying guids_num and step_size):

```
# flint -d /dev/mst/mt4113_pciconf0 q Image type: FS3 FW Version:
10.10.3000 FW Release Date: 29.4.2014 Description: UID GuidNumber
Step Base GUID1: 0002c903002ef500 8 1 Base GUID2:
0002c903002ef508 8 1 Base MAC1: 0002c92ef500 8 1 Base MAC2:
0002c92ef508 8 1 Image VSD: Device VSD: VSD PSID: MT_1240110019
# flint -d /dev/mst/mt4113_pciconf0 -uid 0000000000000001 -ocr sg
guids_num=2 step_size=1 -W- Firmware flash cache access is enabled.
Running in this mode may cause the firmware to hang. Updating GUID
section - OK Updating ITOC section - OK Restoring signature - OK # flint
-d /dev/mst/mt4113_pciconf0 q Image type: FS3 FW Version: 10.10.3000
FW Release Date: 29.4.2014 Description: UID GuidNumber Step Base
GUID1: 0000000000000001 2 1 Orig Base GUID1: 0002c903002ef500 8
1 Base GUID2: 0000000000000003 2 1 Orig Base GUID2:
0002c903002ef508 8 1 Base MAC1: 000000000001 2 1 Orig
Base MAC1: 0002c92ef500 8 1 Base MAC2: 000000000003 2 1
Orig Base MAC2: 0002c92ef508 8 1 Image VSD: Device VSD: VSD
PSID: MT_1240110019
```

3. Change GUIDs for image:

```
# flint -i /tmp/connect-ib.bin q Image type: FS3 FW Version: 10.10.3000
FW Release Date: 29.4.2014 Description: UID GuidNumber Step Base
```

GUID1: 0002c903002ef500 8 1 Base GUID2: 0002c903002ef508 8 1
Base MAC1: 0002c92ef500 8 1 Base MAC2: 0002c92ef508 8 1 Image
VSD: Device VSD: VSD PSID: MT_1240110019 # flint -i /tmp/connect-
ib.bin -uid 000123456abcd sg Updating GUID section - OK Updating
ITOC section - OK Restoring signature - OK # flint -i /tmp/connect-ib.bin
q Image type: FS3 FW Version: 10.10.3000 FW Release Date: 29.4.2014
Description: UID GuidsNumber Step Base GUID1: 000000123456abcd 8
1 Orig Base GUID1: 0002c903002ef500 8 1 Base GUID2:
000000123456abd5 8 1 Orig Base GUID2: 0002c903002ef508 8 1 Base
MAC1: 00000056abcd 8 1 Orig Base MAC1: 0002c92ef500 8 1 Base
MAC2: 00000056abd5 8 1 Orig Base MAC2: 0002c92ef508 8 1 Image
VSD: Device VSD: VSD PSID: MT_1240110019

4. Change GUIDs and MACs for the ConnectX-4 device:

```
# flint -d /dev/mst/mt4115_pciconf0 -guid e41d2d0300570fc0 -mac  
0000e41d2d570fc0 -ocr sg -W- Firmware flash cache access is enabled.  
Running in this mode may cause the firmware to hang. Updating GUID  
section - OK Updating ITOC section - OK Restoring signature - OK # flint  
-d /dev/mst/mt4115_pciconf0 q Image type: FS3 FW Version:  
12.0100.5630 FW Release Date: 23.3.2015 Description: UID  
GuidsNumber Base GUID: e41d2d0300570fc0 4 Base MAC:  
e41d2d570fc0 4 Image VSD: Device VSD: PSID: MT_2190110032 add  
note: GUIDs and MACs can be changed separately on ConnectX4
```

Preparing a Binary Firmware Image for Pre-assembly Burning

In some cases, OEMs may prefer to pre-burn the flash before it is assembled on board. To generate an image for pre-burning for 4th generation (Group I) devices, use the `mlxburn "- striped_image"` flag. The "striped image" file layout is identical to the image layout on the flash, hence making it suitable for burning verbatim. When pre-burning, the GUIDs/MACs inside the image should be unique per device. The following are two methods to pre-burn an image. You can choose the best method suitable for your needs.

Method 1: Pre-burn an Image with Blank GUIDs/MACs

In this method, the image is generated with blank GUIDs and CRCs. The GUIDs are set after the device is assembled using the flint "sg" command. To set

GUIDs take less than 1 second when running on an image with blank GUIDs (through a PCI device).

A device that is burnt with blank GUIDs/MACs will not boot as a functional network device as long as the GUIDs/MACs are not set.

To pre-burn an image with blank GUIDs/MACs:

1. Generate a striped image with blank GUIDs.

```
1 # mlxburn -fw ./fw-ConnectX3-rel.mlx -./MCX354A-FCB_A2-A5.  
2 -I- Generating image ...  
3 -I- Image generation completed successfully.
```

2. Burn the image to a flash using an external burner.
3. (Optional) After assembly, query the image on flash to verify there are no GUIDs on the device.

```
1 # flint -d /dev/mst/mt4099_pci_cr0 q  
2 Image type:      FS2  
3 FW Version:      2.31.5050  
4 FW Release Date: 4.5.2014  
5 Device ID:       4099  
6 Description: Node          Port1          Port2  
7 GUIDs:           ffffffffffffffff ffffffffffffffff ffffffffffff  
8 MACs:            ffffffffffffffff ffffffffffff  
9 VSD:             n/a  
10 PSID:            MT_1090120019  
11  
12  
13 -W- GUIDs/MACs values and their CRC are not set.
```

4. Set the correct GUIDs. Since the image is with blank GUIDs, this operation takes less than 1 second.

```
1 # flint -d /dev/mst/mt4099_pci_cr0 -guid 0x0002c9030abcdef
```

5. Query the image on flash to verify that the GUIDs are set correctly.

```

1  sg# flint -d /dev/mst/mt4099_pci_cr0 q
2  Image type:      FS2
3  FW Version:      2.31.5050
4  FW Release Date: 4.5.2014
5  Device ID:       4099
6  Description:      Node          Port1          Port2
7  GUIDs:            0002c9030abcdef0 0002c9030abcdef1 0002c9030abcdef2
8  MACs:              0002c9030abcdef0 0002c9030abcdef1 0002c9030abcdef2
9  VSD:              n/a
10 PSID:             MT_1090120019

```

Method 2: Pre-burn an Image with Specific GUIDs/MACs for Each Device

In this method, a “base” image is generated with arbitrary default GUIDs and then updated with the correct GUIDs for each device.

To pre-burn an image with specific GUIDs/MACs for each device:

1. Generate the base image with arbitrary default GUIDs.

```

1  # mlxburn -fw./fw-ConnectX3-rel.mlx -c ./MCX354A-FCB_A2-A5

```

2. Per device, set the device specific GUIDs in the image.

```

1  flint -i ./fw-ConnectX3-rel.bin -guid 0x0002c9030abcdef0 -

```

3. (Optional) After assembly, query the image on flash to verify there are no GUIDs on the device.

```

1  sg# flint -i ./fw-ConnectX3-rel.bin -striped_image q
2  Image type:      FS2
3  FW Version:      2.31.5050
4  FW Release Date: 4.5.2014
5  Device ID:       4099
6  Description:      Node          Port1          Port2
7  GUIDs:            0002c9030abcdef0 0002c9030abcdef1 0002c9030abcdef2
8  MACs:              0002c9030abcdef0 0002c9030abcdef1 0002c9030abcdef2
9  VSD:              n/a
10 PSID:             MT_1090120019

```

Now the fw-ConnectX3-rel.bin image can be pre-burned to the flash.
After the assembly, the device would be fully functional.

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